

Mathematics Stage 3 -- Year B -- Unit 40

Integers, Decimals, Percentages & the Cartesian Plane

KLA: Mathematics | Stage 3 | Year 6 | Term: _____ | Duration: 2-3 weeks | Teacher: _____ | Year: _____

Syllabus outcomes

Code	Outcome descriptor
MAO-WM-01	Develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly.
MA3-RN-01	Applies an understanding of place value and the role of zero to represent the properties of numbers.
MA3-RN-03	Determines percentages of quantities, and finds equivalent fractions and decimals for benchmark percentage values.
MA3-AR-01	Selects and applies appropriate strategies to solve addition and subtraction problems.
MA3-MR-01	Selects and applies appropriate strategies to solve multiplication and division problems.
MA3-MR-02	Constructs and completes number sentences involving multiplicative relations, applying the order of operations to calculations.
MA3-GM-01	Locates and describes points on a coordinate plane.

Syllabus content points (addressed through SOLO tracker)

All content points below are covered by students working independently through the SOLO tracker. Mini masterclasses target specific outcomes based on live tracker data.

Red -- Foundation	Yellow -- On-level	Green -- Extension
Decimals (RN-A), Cartesian system (GM-A) + foundational B Compare and order decimals to 3 dp; interpret trailing zeros; estimate decimal size (RN-A). Place negative whole numbers on a number line; negatives from subtraction (RN-B). Write common percentages as fractions and decimals; find 10% of an amount (RN-B). Use place value to multiply and divide a decimal by powers of 10 (MR-B). Grid-map vs number plane; record coordinates in all 4 quadrants (GM-A & B).	Percentages, decimal operations & order of operations (applying) Estimate 10%, 25% and 50% discounts and calculate the sale price. Add and subtract decimals to 3 dp to solve word problems; justify the strategy. Solve multistep word problems requiring more than one operation. Multiply benchmark decimals by single-digit numbers; estimate the product. Use brackets and the order of operations to solve problems.	Extended -- Stage 4 Integers, Percentages & the Cartesian Plane Compare, order, add and subtract positive and negative integers (MA4-INT-C-01). Find percentages of quantities; solve percentage increase and decrease problems (MA4-FRC-C-01). Plot points and linear relationships on the Cartesian plane (MA4-LIN-C-01).

Teaching model

Students work independently through the Word Labs SOLO Tracker (wordlabs.app/solo) across differentiated bands -- Grow (resources), Know (practice), Show (auto-marked assessments). The teacher monitors live dashboard data and pulls small groups for targeted mini masterclasses (15-40 min) based on which outcomes have the most incomplete Show results. Remaining session time is used for incidental one-on-one and small group teaching in response to student questions and tracker data.

Working Mathematically (MAO-WM-01)

Process	How it appears in this unit
Communicating	Students read and write decimals, integers, percentages and coordinates, and explain their strategies.
Understanding and fluency	Students compare/order decimals and integers, find 10%, multiply/divide decimals by powers of 10, and plot coordinates.
Reasoning	Students reason about place value and trailing zeros, why subtraction can give a negative, and which operation order is needed.
Problem solving	Students solve discount, decimal and multistep word problems, and (Stage 4) integer, percentage-change and linear-plotting problems.

Differentiation

Support	HPGE / Extension
Students in this group:	Students in this group:

Pre-test

Date	
Format	
Band place ments / key finding s	

Post-test / unit evaluation

Date	
Key finding s	
Students who need further consol idation	
Next steps	

Resources

See Appendix for Resources

Word Labs SOLO Tracker: wordlabs.app/solo | NSW DoE Stage 3 Year B Unit 40 | NESA Mathematics K-10 Syllabus (2022)

Unit 40 -- Outcome Teaching Record

Find the outcome/s you taught, write your notes, sign and date. Order follows the outcome sequence, not your teaching calendar.

Co de	Outcome	Syllabus content point/s addressed	Mini masterclass teaching notes	Sign	Date
R1	Compare and order decimals to 3 decimal places.	RN-A: compare/order decimals to 3 dp; interpret trailing zeros; approximate decimal size; place on a number line MA3-RN-01 Year A	Mini Lesson 1		
R2	Place integers on a number line.	RN-B: place negative whole numbers on a number line; recognise negatives can result from subtraction MA3-RN-01 Year B	Mini Lesson 2		
R3	Benchmark fractions, decimals, percentages & find 10%.	RN-B: represent common percentages as fractions/decimals; 10% = one-tenth; find 10% of an amount MA3-RN-03 Year B	Mini Lesson 3		
R4	Multiply and divide decimals by powers of 10.	MR-B: use place value to multiply and divide a decimal by powers of 10 MA3-MR-01 Year B	Mini Lesson 4		
R5	The Cartesian coordinate system (4 quadrants).	GM-A & B: grid-map gives an area, the number plane an exact point; record coordinates in all 4 quadrants MA3-GM-01 Year A & B	Mini Lesson 5		
Y1	Estimate discounts and calculate sale prices.	RN-B: estimate 10%/25%/50% discounts; calculate the sale price after a discount MA3-RN-03 Year B	Mini Lesson 6		
Y2	Add and subtract decimals to solve word problems.	AR-B: add/subtract decimals to 3 dp in word problems; justify the strategy MA3-AR-01 Year B	Mini Lesson 7		
Y3	Solve multistep word problems.	AR-B: solve multistep word problems requiring more than one operation MA3-AR-01 Year B	Mini Lesson 8		
Y4	Multiply decimals (mental strategies & estimation).	MR-B: mentally multiply benchmark decimals by single-digit numbers; estimate the product MA3-MR-01 Year B	Mini Lesson 9		
Y5	Order of operations and brackets.	MR-B: agree the order of operations; solve problems using grouping symbols MA3-MR-01, MA3-MR-02 Year B	Mini Lesson 10		
G1	Add, subtract and order integers.	Stage 4 INT: compare/order integers; add and subtract positive and negative integers MA4-INT-C-01 Stage 4 (Core)	Mini Lesson 11		
G2	Percentages of quantities and percentage change.	Stage 4 FRC: find percentages of quantities; solve percentage increase/decrease problems MA4-FRC-C-01 Stage 4 (Core)	Mini Lesson 12		
G3	Plot points and linear relationships on the Cartesian plane.	Stage 4 LIN: plot points of given coordinates; plot linear relationships on the Cartesian plane MA4-LIN-C-01 Stage 4 (Core)	Mini Lesson 13		

Unit 40 -- Mini Lesson Sequence

Integers, Decimals, Percentages & the Cartesian Plane | Stage 3 Year B

Supplement to the SOLO Teacher Program (pages 1-2). Print pages 1-2 only for annotation. Reference these lesson plans when planning mini masterclasses or when a reviewer requires evidence of planned lesson content.

Red band -- Foundation -- Mini Lessons 1-5

Yellow band -- On-level -- Mini Lessons 6-10

Green band -- Extension -- Mini Lessons 11-13

DoE lesson links: Lessons marked DoE Lesson X draw directly from NSW DoE Mathematics Stage 3 Unit 40. Steps have been condensed to 20-40 minutes for mini masterclass delivery. Resources referenced by their original resource numbers from the DoE unit.

Mini Lesson 1 Comparing and ordering decimals [HANDS-ON]

R1 | Red band | DoE Lesson 2 (adapted)

Duration: 25-30 min
Outcomes: R1
Syllabus: MA3-RN-01, MAO-WM-01 | RN-A: compare/order decimals to 3 dp

Linked to: DoE Lesson 2 (adapted) -- Lesson 2 -- Comparing and ordering decimals | Key resources: Decimal number lines, place value charts

Physical materials needed: decimal number lines, place value charts

Learning intention

Students are learning to:
compare and order decimals to 3 decimal places

Success criteria

Students can:

- * I can compare decimals digit by digit using place value.
- * I can order decimals to 3 decimal places on a number line.
- * I know a zero at the end of a decimal (e.g. 0.40) does not change its value.

Mini masterclass structure (25-30 min)

Activate (4 min): Which is bigger, 0.4 or 0.39? Ask students to explain using place value.

Model (10 min): Line up the decimal points; compare tenths, then hundredths, then thousandths. Show $0.4 = 0.40 = 0.400$. Place several decimals on a number line.

Guided practice (10 min): Students order sets of decimals to 3 dp and place them on a number line.

Check (4 min): Exit question: order 0.7, 0.07, 0.77 and 0.707 from smallest to largest.

Key vocabulary

decimal, place value, tenths, hundredths, thousandths, compare, order, number line

Differentiation

Support: Use a place value chart so digits line up under the right column.
Extension: Insert a decimal that lies between two given decimals (e.g. between 0.4 and 0.5).

Assessment

Students compare and order decimals — links to R1 in the SOLO Show task.

Teaching notes (annotate here)

Taught as described / adapted / replaced -- note here:

Mini Lesson 2 Integers on a number line [HANDS-ON]

R2 | Red band | DoE Lesson 1 (adapted)

Duration: 25 min | **Outcomes:** R2 | Syllabus: MA3-RN-01, MAO-WM-01 | RN-B: negatives on a number line

Linked to: DoE Lesson 1 (adapted) -- Lesson 1 -- Locating and representing integers | Key resources: Integer number lines, thermometer model

Physical materials needed: integer number lines

Learning intention		Success criteria	
Students are learning to: place negative whole numbers on a number line		Students can: * I can place negative whole numbers on a number line in relation to zero. * I know numbers get smaller as you move left of zero. * I can explain that subtracting a bigger number from a smaller one gives a negative answer.	
Mini masterclass structure (25 min)			
Activate (3 min): Show a thermometer at −3 degrees. Where is that compared to 0? Model (10 min): Draw a number line through zero; place −1, −2, −3 to the left. Show $2 - 5 = -3$ by counting back past zero. Guided practice (8 min): Students place negative integers on a number line and find results of subtractions that go below zero. Check (4 min): Exit question: place −4 on a number line. What is $1 - 4$?			
Key vocabulary		Differentiation	
integer, negative, positive, zero, number line, less than, subtraction		Support: Use a vertical number line (thermometer) so 'below zero' is concrete. Extension: Order a set of positive and negative integers from smallest to largest.	
Assessment		Teaching notes (annotate here)	
Students place integers on a number line — links to R2 in the SOLO Show task.		Taught as described / adapted / replaced -- note here:	

Mini Lesson 3 Benchmark fractions, decimals, percentages & 10%

R3 | Red band | DoE Lesson 4 (adapted)

Duration: 25-30 min **Outcomes:** R3 Syllabus: MA3-RN-03, MAO-WM-01 | RN-B: benchmark FDP; find 10%

Linked to: DoE Lesson 4 (adapted) -- Lesson 4 -- Connecting fractions, decimals and percentages | Key resources: Hundred grids, FDP cards

Learning intention		Success criteria
Students are learning to: connect benchmark fractions, decimals and percentages and find 10% of an amount		Students can: * I can write common percentages as fractions and decimals (e.g. 50% = $\frac{1}{2}$ = 0.5). * I know 10% is one-tenth of 100%. * I can find 10% of an amount by dividing by 10.
Mini masterclass structure (25-30 min)		
Activate (4 min): What is 50% as a fraction and a decimal? What about 10%? Model (10 min): Build a benchmark table: 50% = $\frac{1}{2}$ = 0.5; 25% = $\frac{1}{4}$ = 0.25; 10% = $\frac{1}{10}$ = 0.1. Find 10% of 80 by dividing by 10 (= 8). Guided practice (10 min): Students write percentages as fractions/decimals and find 10% of amounts. Check (4 min): Exit question: find 10% of 120. Write 25% as a fraction.		
Key vocabulary		Differentiation
percentage, fraction, decimal, benchmark, per cent, one-tenth, of		Support: Use a hundred grid so each percentage is visible. Extension: Use 10% to find 20%, 30% and 5% of an amount.
Assessment		Teaching notes (annotate here)
Students connect FDP and find 10% — links to R3 in the SOLO Show task.		Taught as described / adapted / replaced -- note here:

Mini Lesson 4 Multiply and divide decimals by powers of 10

R4 | Red band | DoE Lesson 6 (adapted)

Duration: 25 min | **Outcomes:** R4 | Syllabus: MA3-MR-01, MAO-WM-01 | MR-B: $\times/\div$ decimals by powers of 10

Linked to: DoE Lesson 6 (adapted) -- Lesson 6 -- Multiplying and dividing decimals | Key resources: Place value charts

Learning intention		Success criteria
Students are learning to: use place value to multiply and divide a decimal by powers of 10		Students can: * I know multiplying by 10 makes each digit 10 times bigger (shifts left). * I know dividing by 10 makes each digit 10 times smaller (shifts right). * I can multiply and divide decimals by 10, 100 and 1000.
Mini masterclass structure (25 min)		
Activate (3 min): What is 3.4×10? Ask students to explain what happens to the digits. Model (10 min): On a place value chart, $\times 10$ shifts digits one place left ($3.4 \rightarrow 34$); $\div 10$ shifts one place right ($3.4 \rightarrow 0.34$). Extend to $\times 100$ and $\div 100$. Guided practice (8 min): Students multiply and divide decimals by 10, 100 and 1000 using place value. Check (4 min): Exit question: $2.5 \times 100 = ?$ and $47 \div 10 = ?$		
Key vocabulary		Differentiation
place value, multiply, divide, powers of 10, shift, tenths, hundredths		Support: Use a place value chart and physically move the digits. Extension: Multiply and divide by 1000, and explain the pattern in the number of place shifts.
Assessment		Teaching notes (annotate here)
Students $\times/\div$ decimals by powers of 10 — links to R4 in the SOLO Show task.		Taught as described / adapted / replaced -- note here:

Mini Lesson 5 The Cartesian coordinate system [HANDS-ON]

R5 | Red band | DoE Lesson 8 (adapted)

Duration: 30 min | **Outcomes:** R5 | Syllabus: MA3-GM-01, MAO-WM-01 | GM-A & B: grid-map vs number plane; 4 quadrants

Linked to: DoE Lesson 8 (adapted) -- Lesson 8 -- The Cartesian coordinate system | Key resources: Coordinate grids (4 quadrants), counters

Physical materials needed: coordinate grids, counters

Learning intention	Success criteria
Students are learning to: read and record coordinates of points in all four quadrants	Students can: * I know the number plane has an x-axis and a y-axis that cross at the origin (0,0). * I can write a point's coordinates as (x, y), going across then up/down. * I can read and plot points in all four quadrants, including negatives.

Mini masterclass structure (30 min)

Activate (4 min): Compare a grid-map reference (an area, like B3) with a number-plane point (an exact spot, like (2,3)).

Model (12 min): Draw the four quadrants; mark the origin. Plot (2,3), (-2,3), (-2,-3) and (2,-3). Always go across (x) first, then up/down (y).

Guided practice (10 min): Students read and plot points in all four quadrants on a coordinate grid.

Check (4 min): Exit question: write the coordinates of a point 3 left and 2 down from the origin.

Key vocabulary	Differentiation
Cartesian plane, x-axis, y-axis, origin, coordinates, quadrant, ordered pair	Support: Provide a labelled grid and remind 'across before up'. Extension: Plot points to draw a simple shape and name its vertices' coordinates.
Assessment	Teaching notes (annotate here)
Students read and record coordinates — links to R5 in the SOLO Show task.	Taught as described / adapted / replaced -- note here:

Mini Lesson 6 Percentage discounts

Y1 | Yellow band | DoE Lesson 5 (adapted)

Duration: 30 min | **Outcomes:** Y1 | Syllabus: MA3-RN-03, MAO-WM-01 | RN-B: estimate discounts; sale price

Linked to: DoE Lesson 5 (adapted) -- Lesson 5 -- Estimating percentage discounts | Key resources: Catalogue/price cards (AUD)

Learning intention		Success criteria
Students are learning to: estimate a discount and calculate the sale price of an item		Students can: * I can find 10%, 25% and 50% of a price. * I can subtract the discount from the original price to get the sale price. * I can estimate a discount mentally to check my answer.
Mini masterclass structure (30 min)		
Activate (4 min): A \$80 jacket is 25% off. Roughly how much will you save? Model (12 min): 25% of \$80 = \$20, so sale price = \$80 – \$20 = \$60. Show 50% = half, 10% = ÷10. Estimate first, then calculate. Guided practice (10 min): Students calculate sale prices after 10%, 25% and 50% discounts (AUD). Check (4 min): Exit question: a \$40 game is 10% off. What is the sale price?		
Key vocabulary		Differentiation
percentage, discount, sale price, estimate, original price, per cent, AUD		Support: Provide the discount amount so students focus on subtracting from the price. Extension: Find the discount when 20% or 75% off, using 10% and 25% as building blocks.
Assessment		Teaching notes (annotate here)
Students calculate discounts and sale prices — links to Y1 in the SOLO Show task.		Taught as described / adapted / replaced -- note here:

Mini Lesson 7 Adding and subtracting decimals

Y2 | Yellow band | DoE Lesson 3 (adapted)

Duration: 30 min | **Outcomes:** Y2 | Syllabus: MA3-AR-01, MAO-WM-01 | AR-B: add/subtract decimals; justify

Linked to: DoE Lesson 3 (adapted) -- Lesson 3 -- Strategies to solve addition and subtraction problems | Key resources: Number lines, place value charts

Learning intention		Success criteria
Students are learning to: add and subtract decimals to 3 decimal places to solve word problems		Students can: * I can line up the decimal points to add or subtract decimals. * I can solve word problems involving adding and subtracting decimals. * I can explain why my strategy works.
Mini masterclass structure (30 min)		
Activate (4 min): Buy items for \$3.50 and \$2.75. How much altogether? Model (12 min): Line up the decimal points; add place by place (\$6.25). Subtract change from \$10. Model adding/subtracting on a number line, and justify the strategy. Guided practice (10 min): Students solve add/subtract decimal word problems and explain their strategy. Check (4 min): Exit question: $7.5 - 3.25 = ?$ and explain how you did it.		
Key vocabulary		Differentiation
decimal, add, subtract, line up, decimal point, strategy, justify, word problem		Support: Use a place value chart so the decimal points and columns line up. Extension: Solve a two-step money problem mixing addition and subtraction of decimals.
Assessment		Teaching notes (annotate here)
Students add and subtract decimals — links to Y2 in the SOLO Show task.		Taught as described / adapted / replaced -- note here:

Mini Lesson 8 Multistep word problems

Y3 | Yellow band | DoE Lesson 3 (adapted)

Duration: 30 min | **Outcomes:** Y3 | Syllabus: MA3-AR-01, MAO-WM-01 | AR-B: multistep word problems

Linked to: DoE Lesson 3 (adapted) -- Lesson 3 -- Strategies to solve addition and subtraction problems | Key resources: Word problem cards

Learning intention		Success criteria
Students are learning to: solve multistep word problems requiring more than one operation		Students can: * I can break a problem into steps and decide which operation each step needs. * I can carry out the steps in order to reach the answer. * I can check my answer is reasonable.
Mini masterclass structure (30 min)		
Activate (4 min): Read a 2-step problem aloud. Ask: what do we work out first? Model (12 min): Model a multistep problem: e.g. buy 3 items at \$4.50, pay with \$20 — first multiply ($3 \times 4.50 = 13.50$), then subtract ($20 - 13.50 = 6.50$). Guided practice (10 min): Students solve multistep word problems, showing each step. Check (4 min): Exit question: a 2-step problem with addition then subtraction.		
Key vocabulary		Differentiation
multistep, operation, step, multiply, add, subtract, estimate, reasonable		Support: Provide a step-by-step planning frame (Step 1 ..., Step 2 ...). Extension: Write a multistep problem of your own and solve it.
Assessment		Teaching notes (annotate here)
Students solve multistep word problems — links to Y3 in the SOLO Show task.		Taught as described / adapted / replaced -- note here:

Mini Lesson 9 Multiplying decimals (mental & estimation)

Y4 | Yellow band | DoE Lesson 6 (adapted)

Duration: 25-30 min **Outcomes:** Y4 Syllabus: MA3-MR-01, MAO-WM-01 | MR-B: multiply decimals; estimate

Linked to: DoE Lesson 6 (adapted) -- Lesson 6 -- Multiplying and dividing decimals | Key resources: Place value charts

Learning intention

Students are learning to:
multiply a decimal by a single-digit number and estimate the product

Success criteria

Students can:

- * I can multiply a benchmark decimal (like 0.5 or 2.5) by a single-digit number mentally.
- * I can estimate a decimal $\times$ whole number by rounding.
- * I can check whether a calculator answer is the right size.

Mini masterclass structure (25-30 min)

Activate (3 min): What is 0.5×6 ? Ask students to explain (half of 6).

Model (12 min): $0.5 \times 6 = 3$; $2.5 \times 4 = 10$. Estimate 3.8×5 by rounding to $4 \times 5 = 20$, then compare with the exact 19.0.

Guided practice (10 min): Students multiply benchmark decimals by single-digit numbers and estimate products.

Check (4 min): Exit question: estimate 4.2×3 , then work out the exact answer.

Key vocabulary

decimal, multiply, product, estimate, round, benchmark, magnitude

Differentiation

Support: Start with 0.5 (half) and 0.25 (quarter) of single-digit numbers.
Extension: Multiply a 2-decimal-place number by a single digit and estimate first.

Assessment

Students multiply decimals and estimate — links to Y4 in the SOLO Show task.

Teaching notes (annotate here)

Taught as described / adapted / replaced -- note here:

Mini Lesson 10 Order of operations and brackets

Y5 | Yellow band | DoE Lesson 7 (adapted)

Duration: 30 min | **Outcomes:** Y5 | Syllabus: MA3-MR-01, MA3-MR-02, MAO-WM-01 | MR-B: order of operations; grouping symbols

Linked to: DoE Lesson 7 (adapted) -- Lesson 7 -- Brackets and the order of operations | Key resources: Number sentence cards

Learning intention		Success criteria
Students are learning to: use brackets and the order of operations to solve number problems		Students can: * I know we agree on an order so everyone gets the same answer. * I can do brackets first, then multiplication/division, then addition/subtraction. * I can solve problems that use grouping symbols.
Mini masterclass structure (30 min)		
Activate (4 min): Work out $2 + 3 \times 4$ two different ways. Why do we need an agreed order? Model (12 min): Order of operations: brackets, then $\times$ and $\div$, then $+$ and $-$. So $2 + 3 \times 4 = 14$, but $(2 + 3) \times 4 = 20$. Brackets change the answer. Guided practice (10 min): Students evaluate number sentences with and without brackets. Check (4 min): Exit question: work out $10 - (2 + 3) \times 1$ and $20 \div (5 - 1)$.		
Key vocabulary		Differentiation
order of operations, brackets, grouping symbols, multiply, divide, add, subtract		Support: Provide the operation order as a checklist to follow. Extension: Place brackets in a number sentence to make it equal a given target.
Assessment		Teaching notes (annotate here)
Students apply the order of operations — links to Y5 in the SOLO Show task.		Taught as described / adapted / replaced -- note here:

Mini Lesson 11 Adding, subtracting and ordering integers

G1 | Green band | Original mini lesson (Stage 4 extension)

Duration: 30 min | **Outcomes:** G1 | Syllabus: MA4-INT-C-01, MAO-WM-01 | Stage 4: compare/order/add/subtract integers

Learning intention		Success criteria
Students are learning to: compare, order, add and subtract positive and negative integers		Students can: * I can compare and order integers using < and >. * I can add a negative integer (move left on the number line). * I can subtract to cross zero into negative numbers.
Mini masterclass structure (30 min)		
Activate (4 min): Which is greater, -5 or -2? Use a number line to decide. Model (12 min): On a number line: $3 + (-5) = -2$ (move left 5); $-1 - 3 = -4$. Order a set like $-4, 2, -1, 3$. Guided practice (10 min): Students compare, order, add and subtract positive and negative integers. Check (4 min): Exit question: work out $-3 + 5$ and $2 - 6$.		
Key vocabulary		Differentiation
integer, positive, negative, less than, greater than, add, subtract, number line		Support: Use a number line for every calculation to make direction concrete. Extension: Solve a real-life integer problem (temperature change or money owed).
Assessment		Teaching notes (annotate here)
Students work with integers — extends R2.		Taught as described / adapted / replaced -- note here:

Mini Lesson 12 Percentages of quantities and percentage change

G2 | Green band | Original mini lesson (Stage 4 extension)

Duration: 30 min | **Outcomes:** G2 | Syllabus: MA4-FRC-C-01, MAO-WM-01 | Stage 4: percentages of quantities; % change

Learning intention		Success criteria	
Students are learning to: find a percentage of a quantity and work out a percentage increase or decrease		Students can: * I can find any percentage of a quantity (e.g. 15% of 60). * I can increase or decrease a quantity by a percentage. * I can work out the percentage change between two amounts.	
Mini masterclass structure (30 min)			
Activate (4 min): Find 15% of 60 by combining 10% (6) and 5% (3). Model (12 min): 15% of 60 = 9. Increase \$50 by 20%: 20% of 50 = 10, so \$60. Decrease by a percentage similarly. Guided practice (10 min): Students find percentages of quantities and solve percentage increase/decrease problems. Check (4 min): Exit question: increase 80 by 25%.			
Key vocabulary		Differentiation	
percentage, quantity, of, increase, decrease, percentage change, per cent		Support: Build any percentage from 10%, 5% and 1% chunks. Extension: Find the percentage change when a price rises from \$40 to \$50.	
Assessment		Teaching notes (annotate here)	
Students find percentages of quantities and change — extends R3/Y1.		Taught as described / adapted / replaced -- note here:	

Mini Lesson 13		Points and linear relationships on the Cartesian plane	
G3 Green band Original mini lesson (Stage 4 extension)			
Duration: 30 min		Outcomes: G3	
Syllabus: MA4-LIN-C-01, MAO-WM-01 Stage 4: plot points; plot linear relationships			
Learning intention		Success criteria	
Students are learning to: plot points and a simple linear relationship on the Cartesian plane		Students can: * I can plot points of given coordinates in all four quadrants. * I can complete a table of values for a simple rule (e.g. $y = x + 2$). * I can plot the points from the table and join them to make a straight line.	
Mini masterclass structure (30 min)			
Activate (4 min): Plot (1,3), (2,4), (3,5). What do you notice about the points?			
Model (12 min): For $y = x + 2$, build a table: $x = 0,1,2,3 \rightarrow y = 2,3,4,5$. Plot the points; they line up — join them into a straight line.			
Guided practice (10 min): Students complete a table of values for a simple rule and plot the line.			
Check (4 min): Exit question: complete the table for $y = 2x$ and plot the points.			
Key vocabulary		Differentiation	
Cartesian plane, coordinates, table of values, rule, linear, straight line, plot		Support: Provide the table partly completed so students plot the points. Extension: Find the rule that produces a given straight line of points.	
Assessment		Teaching notes (annotate here)	
Students plot points and a linear relationship — extends R5.		Taught as described / adapted / replaced -- note here:	

Unit 40 -- Resource Appendix

Every link verified. Videos open in a browser; worksheets are print-ready PDFs (see the download checklist at the end).

Type	Resource	Link
R1 -- Compare and order decimals to 3 decimal places.		
Video	Comparing and Ordering Decimals Math with Mr. J	https://www.youtube.com/watch?v=KWslYjXKhrM
Video	How to Order Decimals Math with Mr. J	https://www.youtube.com/watch?v=1dfedgVxA54
Worksheet (print)	Corbettmaths — Ordering Decimals (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/ordering-decimals-pdf.pdf
R2 -- Place integers on a number line.		
Video	Integers on a Number Line — Missing Numbers Math with Mr. J	https://www.youtube.com/watch?v=IKZp-jEvGqM
Video	How to Compare and Order Integers (Positives & Negatives) Math with Mr. J	https://www.youtube.com/watch?v=2FbafvAGhZ0
Worksheet (print)	Corbettmaths — Negative Numbers (PDF)	https://corbettmaths.com/wp-content/uploads/2018/11/Negative-Numbers.pdf
R3 -- Benchmark fractions, decimals, percentages & find 10%.		
Video	How to Find a Percent of a Number Math with Mr. J	https://www.youtube.com/watch?v=5CAncQ4TZD4
Video	Converting Between Fractions / Decimals / Percents — Mini Course Math with Mr. J	https://www.youtube.com/watch?v=2vjbNZaFz3c
Worksheet (print)	Corbettmaths — Fractions / Decimals / Percentages (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/fractions-decimals-percentages-pdf.pdf
R4 -- Multiply and divide decimals by powers of 10.		
Video	Multiplying and Dividing by Powers of 10 Math with Mr. J	https://www.youtube.com/watch?v=HH-yrNS80Cg
Video	Dividing by Powers of Ten Math with Mr. J	https://www.youtube.com/watch?v=E8b5pDvt2AE
Worksheet (print)	Corbettmaths — Place Value (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/place-value-pdf.pdf
R5 -- The Cartesian coordinate system (4 quadrants).		
Video	How to Plot Points on a Coordinate Plane — Positive and Negative Coordinates Math with Mr. J	https://www.youtube.com/watch?v=5NyamsbLMo
Video	Quadrants of the Coordinate Plane — What are the Four Quadrants? Math with Mr. J	https://www.youtube.com/watch?v=uc9hWu73Phw
Worksheet (print)	Corbettmaths — Coordinates (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/coordinates-pdf1.pdf
Y1 -- Estimate discounts and calculate sale prices.		
Video	How to Calculate a Discount and Sale Price Math with Mr. J	https://www.youtube.com/watch?v=5qxX222sLiE
Video	How to Calculate a Discount without a Calculator Math with Mr. J	https://www.youtube.com/watch?v=og9bDI2XJfA
Worksheet (print)	Corbettmaths — Percentage of an Amount (Non-Calculator) (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/percentage-of-an-amount-non-calculator-pdf1.pdf
Y2 -- Add and subtract decimals to solve word problems.		
Video	Adding and Subtracting Decimals (How to) Math with Mr. J	https://www.youtube.com/watch?v=PnwLv6khwk8
Video	How to Add and Subtract Decimals — Step-by-Step Examples Math with Mr. J	https://www.youtube.com/watch?v=Cg-TeiaSa8
Worksheet (print)	Corbettmaths — Ordering Decimals (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/ordering-decimals-pdf.pdf
Y3 -- Solve multistep word problems.		
Video	Multi-Step Problems With All Operations (Road Trip Math) Easy Math with Mrs. Easley	https://www.youtube.com/watch?v=z0LbNsZN71w
Worksheet (print)	Corbettmaths — Order of Operations (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/order-of-operations-pdf.pdf
Worksheet (print)	Corbettmaths — Place Value (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/place-value-pdf.pdf
Y4 -- Multiply decimals (mental strategies & estimation).		

Type	Resource	Link
Video	Multiply a Whole Number by a Decimal Math with Mr. J	https://www.youtube.com/watch?v=tsOibhsgYoQ
Video	How to Multiply a Whole Number and a Decimal Math with Mr. J	https://www.youtube.com/watch?v=FbZr5br9ZOg
Worksheet (print)	Corbettmaths — Multiplying and Dividing Decimals (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/decimals-multiplying-and-dividing.pdf
Y5 -- Order of operations and brackets.		
Video	Order of Operations — A Step-By-Step Guide PEMDAS Math with Mr. J	https://www.youtube.com/watch?v=x41BCrYh8Kc
Video	Order of Operations — Parentheses / Brackets / Braces PEMDAS Math with Mr. J	https://www.youtube.com/watch?v=u7Lack4Tmx8
Worksheet (print)	Corbettmaths — Order of Operations (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/order-of-operations-pdf.pdf
G1 -- Add, subtract and order integers.		
Video	How to Add Negative Numbers (Integers) Math with Mr. J	https://www.youtube.com/watch?v=1VFj_coqKQU
Video	Subtracting Integers Using a Number Line Math with Mr. J	https://www.youtube.com/watch?v=Dfytkh_IYME
Worksheet (print)	Corbettmaths — Negative Numbers (PDF)	https://corbettmaths.com/wp-content/uploads/2018/11/Negative-Numbers.pdf
G2 -- Percentages of quantities and percentage change.		
Video	Percent Change — A Step-By-Step Guide Math with Mr. J	https://www.youtube.com/watch?v=ZL2ILv98IZU
Video	How to Find a Percent of a Number Math with Mr. J	https://www.youtube.com/watch?v=5CAncQ4TZD4
Worksheet (print)	Corbettmaths — Percentage Change (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/percentage-change-pdf.pdf
G3 -- Plot points and linear relationships on the Cartesian plane.		
Video	Plotting Points on a Coordinate Plane — All 4 Quadrants Math with Mr. J	https://www.youtube.com/watch?v=pl9nSVzRWvA
Video	Graphing Linear Equations (using a Table of Values) Math Help Videos	https://www.youtube.com/watch?v=ct1dRi7IcQw
Worksheet (print)	Corbettmaths — Drawing Linear Graphs (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/drawing-linear-graphs-pdf.pdf
Beyond SOLO -- Stage 4 extension		
Video	How to Add Negative Numbers (Integers) Math with Mr. J	https://www.youtube.com/watch?v=1VFj_coqKQU
Video	Percent Change — A Step-By-Step Guide Math with Mr. J	https://www.youtube.com/watch?v=ZL2ILv98IZU
Video	Graphing Linear Equations (using a Table of Values) Math Help Videos	https://www.youtube.com/watch?v=ct1dRi7IcQw
Worksheet (print)	Corbettmaths — Negative Numbers (PDF)	https://corbettmaths.com/wp-content/uploads/2018/11/Negative-Numbers.pdf
Worksheet (print)	Corbettmaths — Drawing Linear Graphs (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/drawing-linear-graphs-pdf.pdf

Worksheet download checklist

Download and print these PDFs before the unit (or save to your class drive). Tick each as you go.

- ☐ Corbettmaths — Ordering Decimals (PDF) -- <https://corbettmaths.com/wp-content/uploads/2013/02/ordering-decimals-pdf.pdf>
- ☐ Corbettmaths — Negative Numbers (PDF) -- <https://corbettmaths.com/wp-content/uploads/2018/11/Negative-Numbers.pdf>
- ☐ Corbettmaths — Fractions / Decimals / Percentages (PDF) -- <https://corbettmaths.com/wp-content/uploads/2013/02/fractions-decimals-percentages-pdf.pdf>
- ☐ Corbettmaths — Place Value (PDF) -- <https://corbettmaths.com/wp-content/uploads/2013/02/place-value-pdf.pdf>
- ☐ Corbettmaths — Coordinates (PDF) -- <https://corbettmaths.com/wp-content/uploads/2013/02/coordinates-pdf1.pdf>
- ☐ Corbettmaths — Percentage of an Amount (Non-Calculator) (PDF) -- <https://corbettmaths.com/wp-content/uploads/2013/02/percentage-of-an-amount-non-calculator-pdf1.pdf>
- ☐ Corbettmaths — Order of Operations (PDF) -- <https://corbettmaths.com/wp-content/uploads/2013/02/order-of-operations-pdf.pdf>
- ☐ Corbettmaths — Multiplying and Dividing Decimals (PDF) -- <https://corbettmaths.com/wp-content/uploads/2013/02/decimals-multiplying-and-dividing.pdf>
- ☐ Corbettmaths — Percentage Change (PDF) -- <https://corbettmaths.com/wp-content/uploads/2013/02/percentage-change-pdf.pdf>
- ☐ Corbettmaths — Drawing Linear Graphs (PDF) -- <https://corbettmaths.com/wp-content/uploads/2013/02/drawing-linear-graphs-pdf.pdf>